

CA-NAV — Computer Architecture (CS-328) — Master Index

Weeks 1–15 · 17 Aug – 29 Nov 2026 · **Final 60** / **Sessional 40** · load 5.0 · difficulty 5/5

Open this file first. It is the map: where every topic lives, every line offset, every resource. **Three-file pack** (all in `terms/2026-fall/CA/`): | **File** | **Holds** | `---|---` | `CA-NAV.md` | This map — glance tables, master offsets, resources index | | `CA-BEFORE-MID.md` | **Weeks 1–8** (midterm) — full per-week content, chapters 1–4 | | `CA-AFTER-MID.md` | **Weeks 9–15** (final) — full per-week content, chapter 5 |

Where to look (fast answers)

I want...	Open
A topic with its defs/formulas/numericals	that week's block in BEFORE-MID / AFTER-MID
Any week's exact line offsets	Master offset table below
A full fear-killer pack (verbatim)	that week's block — packs carry per-question offsets
Chapter breakdown (ISA / hazards / cache-VM)	the week that bundles it (mapping below)
Books / YT playlists / MOOCs / practice banks	Resources section in this file
Exam-day order	Exam stacks section
Which official module covers what	Syllabus M1–M9 section

Course facts

- **Code:** CS-328 · **Credits:** 3+0 (theory only) · **Contact:** 3 lectures/week
- **Weightage:** Final 60% / Sessional 40% (web-verified NED policy)
- **Difficulty:** 5/5 · **GPA Risk:** High
- **Midterm:** Week 8 (05–11 Oct) · **Final:** Week 15 (23–29 Nov)
- **Examinability note:** official syllabus marks ISA/single-cycle/cache/TLB as *background* — *not examinable*, but narrative + exam-probability treat cache/TLB as examinable. ISA/single-cycle scheduled P2. Confirm with professor.

Master weekwise line-offset table

Offsets are line numbers into the named source file. `Fear-Killer-Packs.md` and `Week-by-Week-Narrative.md` are the two anchored sources; `weeks/CA-Wn.md` files live in `weeks/`.

Week	Narrative	Pack heading	Pack questions	weeks/CA-Wn	Chapter
W1	L17–30	L13 (CA-W1)	res L14 · Q1 L16 · Q2 L17 · Q3 L18 · Q4 L19	CA-W1 L1–46	01 (L1–178) + 02 (L1–52)
W2	L33–46	L21 (CA-W2)	res L22 · Q1 L24 · Q2 L25 · Q3 L26	CA-W2 L1–44	03 (L1–46)
W3	L49–64	L28 (CA-W3)	res L29 · Q1 L31 · Q2 L32 · Q3 L33	CA-W3 L1–43	03
W4	L68–81	L35 (CA-W4)	res L36 · Q1 L38 · Q2 L39 · Q3 L40	CA-W4 L1–42	04 (L1–57)
W5	L85–100	L42 (CA-W5)	res L43 · Q1 L45 · Q2 L46 · Q3 L47 · Q4 L48 · Q5 L49	CA-W5 L1–43	04
W6	L104–117	L51 (CA-W6)	res L52 · Q1 L54 · Q2 L55 · Q3 L56 · Q4 L57 · Q5 L58	CA-W6 L1–43	04
W7	L121–137	L60 (CA-W7)	res L61 · Q1 L63 · Q2 L64 · Q3 L65 · Q4 L66	CA-W7 L1–42	01–04
W8	L141–149	L68 (no pack)	—	CA-W8 L1–29	—

W9	L152–171	L71 (CA-W9)	res L72 · Q1 L74 · Q2 L75 · Q3 L76 · Q4 L77	CA-W9 L1–41	—
W10	L175–189	L79 (CA-W10)	res L80 · Q1 L82 · Q2 L83 · Q3 L84 · Q4 L85 · Q5 L86	CA-W10 L1–44	—
W11	L193–207	L88 (CA-W11)	res L89 · Q1 L91 · Q2 L92 · Q3 L93	CA-W11 L1–43	05 (cache)
W12	L211–228	L95 (CA-W12)	res L96 · Q1 L98 · Q2 L99 · Q3 L100	CA-W12 L1–42	05 (VM)
W13	L232–246	L102 (CA-W13)	res L103 · Q1 L105 · Q2 L106 · Q3 L107	CA-W13 L1–43	—
W14	L250–269	L109 (CA-W14)	res L110 · Q1 L112 · Q2 L113 · Q3 L114 · Q4 L115 · Q5 L116	CA-W14 L1–47	01–05
W15	L273–283	L118 (no pack)	—	(W15 note at CA-W14 L47)	—

Syllabus M1–M9 → where it lives

Module	Hours	Chapter breakdown?	Covered in
M1 Multi-Cycle Processor Design	6	✓ Ch3	W2–W3
M2 Control Unit — FSM + Microprogrammed	6	✗ pack-level	W3–W4
M3 ILP Concepts + Pipeline Hazards	8	△ Ch4 (hazards only)	W4–W6 (ILP at W9)
M4 Forwarding + Hazard Detection	4	✓ Ch4	W5–W6
M5 Compiler ILP + Branch Prediction	4	✗ pack-level	W6, W9
M6 Multiple Issue (VLIW/Superscalar)	4	✗ pack-level	W10
M7 SIMD & GPU	4	✗ pack-level	W10
M8 Loop-Level Parallelism	4	✗ pack-level	W9
M9 Memory System / VM / RAID	8	△ Ch5 (cache+VM; RAID pack-only)	W11–W13
Background (ISA, SC datapath, cache/TLB, I/O)	—	✓ Ch1, Ch2, Ch5	W1, W11–W13 (P2)

Gaps are flagged, not fabricated: M2/M5/M6/M7/M8 have no chapter breakdowns — mapped at pack level (per `weeks/README.md` resolution).

Exam probability table (editorial — study prioritization only)

Topic	Midterm	Final	Numerical	Diagram	Theory
MIPS ISA	90%	40%	20%	30%	30%
SC datapath	85%	30%	10%	50%	25%
Multi-cycle	80%	40%	15%	40%	30%
FSM control	70%	40%	10%	45%	30%
Pipeline hazards	60%	90%	30%	30%	25%
Branch prediction	40%	70%	25%	25%	30%
ILP/VLIW	0%	60%	25%	15%	40%
Cache	0%	80%	35%	20%	30%
Virtual memory	0%	75%	35%	20%	30%
RAID	0%	50%	20%	20%	40%

Marks allocation

Component	Weight	Strategy
Pipeline hazard tracing	~20%	Drill timing diagrams until reflex
Multi-cycle datapath	~15%	Draw from memory daily
Cache/VM numericals	~15%	Formula mastery
FSM control design	~10%	State diagram practice
Theory/definitions	~40%	Anki flashcards

Exam stacks

- **Midterm (W8, 05–11 Oct):** past paper 60 min → blank page 30 min → error log 20 min. Answer diagram/numerical first, theory second. Check every control signal and hazard classification. Sleep 8 h. Ledger frozen.
- **Final (W15, 23–29 Nov):** numericals → diagrams → theory. Label every forwarding path. Flag stuck problems and move on. Sleep 9 h (banked from W12).

Resources

YouTube playlists

Playlist	Channel	Videos	Why
Basic Computer Architecture — Lecture Videos	Smruti R. Sarangi (IIT Delhi)	48	Official lecture series; matches NPTEL noc25_cs83; ch 8–12 = processor, pipelining, memory, multiprocessor/GPU, I/O
CS61C: Great Ideas in Computer Architecture	UC Berkeley	~40+	Top-tier; digital logic → parallel processing
Build a Superscalar CPU	Independent creator	~10+	Hands-on superscalar core in simulation (W10)

Textbooks

- **Hennessy & Patterson** — *Computer Architecture: A Quantitative Approach*, 7th Ed (2025). ISBN **978-0-443-15406-5**. Gold standard — ILP, VLIW, superscalar, GPU, memory hierarchy, VM.
- **Patterson & Hennessy** — *Computer Organization and Design: The Hardware/Software Interface* (RISC-V Ed, 2nd ed). ISBN **978-0-12-820331-6**. Ch 4 = processor/pipelining/hazards, Ch 5 = memory hierarchy + RAID (5.11), App C = microprogrammed control.
- **Smruti R. Sarangi** — *Basic Computer Architecture*, 2nd Ed. Per-week chapters listed in the narrative's Resources lines.

Free MOOCs

Course	Source	Notes
NPTEL: Computer Architecture	Prof. Smruti Sarangi, IIT Delhi	12 weeks; matches Sarangi playlist
MIT OCW: 6.823 Computer System Architecture	MIT	Full lecture notes + assignments
Coursera: Computer Architecture	Princeton University	Self-paced

Problem banks / practice

Resource	Link
UC Berkeley CS152 Homeworks	https://inst.eecs.berkeley.edu/~cs152/sp26/
MIT 6.823 Past Quizzes	https://csg.csail.mit.edu/6.823S21/
GeeksforGeeks CA Quiz	https://www.geeksforgeeks.org/quizzes/computer-organization-and-architecture-gg/ (UNVERIFIED)
H&P End-of-Chapter Exercises	https://shop.elsevier.com/books/computer-architecture/hennessy/978-0-443-15406-5

Resource	Link
Cheatography: Computer Architecture	https://cheatography.com/sheetocheat/cheat-sheets/computer-architecture/
EduRev: CAO Cheatsheet	https://edurev.in/t/488984/Cheatsheet-Computer-Architecture-Organisation--CAO-
Codecademy: CA Cheatsheet	https://www.codecademy.com/learn/computer-architecture/modules/intro-to-computer-architecture/cheatsheet
Roger Fan: CA Cheat Sheet	https://roger.tw/notes/University-Notes/Computer-Architecture/Computer-Architecture-(Cheat-Sheet)

Secret weapon

gem5 Simulator — <https://www.gem5.org> — cycle-level CPU modeling, cache hierarchies, branch predictors, IPC/CPI. Free, 1600+ GitHub forks.

Where-everything-lives index (source → embedded location)

Source file	~Lines	Embedded in
01-Course-Overview.md	89	NAV facts/weights + per-unit headers in both files
02-Official-Syllabus.md	75	NAV (M1–M9 map + textbooks)
03-Chapter-Breakdowns/01 ISA	178	BEFORE-MID W1 (verbatim)
03-Chapter-Breakdowns/02 Single-cycle	52	BEFORE-MID W1 (verbatim)
03-Chapter-Breakdowns/03 Multi-cycle	46	BEFORE-MID W2/W3 (verbatim)
03-Chapter-Breakdowns/04 Hazards	57	BEFORE-MID W4–W6 (verbatim)
03-Chapter-Breakdowns/05 Cache/VM	45	AFTER-MID W11/W12 (verbatim)
Definition-Book.md	44	split into per-week "Definitions" blocks (verbatim + L-numbers)
Formula-Book.md	29	split into per-week "Formulas" blocks (verbatim + L-numbers)
Diagram-Book.md	26	per-week "Diagrams" numbered refs
Numerical-Book.md	39	per-week "Numericals" numbered types
Top-100-Questions.md	116	per-week ranges by bucket
Top-10-Tricky-Concepts.md	21	per-week items (full list #1–#10 in both files' revision weeks)
Marks-Allocation.md	12	NAV (this file)
Resources.md	45	NAV (this file) + per-week book/video lines
Fear-Killer-Packs.md	121	per-week packs (verbatim, with per-question offsets)
Week-by-Week-Narrative.md	291	per-week narrative (verbatim)
weeks/CA-W1...W14 + README	~650	per-week anchors, P0 floor, drill targets
categorical-analysis.md	69	NOT embedded — superseded/historical (2026-08-04)

Generated from the CA pack (CS-328). Every offset verified against CA-TOC.md, weeks/README.md, and the source files themselves.

Schedule at a glance

W	Dates	Variant	File	Topics	Chapter
1	17–23 Aug	P2	BEFORE	MIPS ISA & single-cycle intro	01 + 02
2	24–30 Aug	P2	BEFORE	Single-cycle datapath	03

3	31 Aug–06 Sep	P2	BEFORE	Multi-cycle FSM control	03
4	07–13 Sep	P0	BEFORE	Microprogramming & pipeline intro	04
5	14–20 Sep	P0	BEFORE	Pipeline hazards I (structural/data)	04
6	21–27 Sep	P0	BEFORE	Pipeline hazards II (detection/branch pred)	04
7	28 Sep–04 Oct	P1	BEFORE	Midterm revision	01–04
8	05–11 Oct	MIDTERM	BEFORE	Exam week — no new material	—
9	12–18 Oct	P1	AFTER	ILP & compiler techniques (recovery)	— (pack-level)
10	19–25 Oct	P1	AFTER	VLIW, superscalar, SIMD/GPU	— (pack-level)
11	26 Oct–01 Nov	P0	AFTER	Memory hierarchy & cache design	05
12	02–08 Nov	P0	AFTER	Virtual memory & TLB	05
13	09–15 Nov	P1	AFTER	RAID & advanced topics	— (pack-level)
14	16–22 Nov	P0	AFTER	Final taper — past paper FIRST	01–05
15	23–29 Nov	FINAL	AFTER	Exam — execute	—

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week.